

Troubleshooting Cables and Physical Layer — Guided Notes

Unit 5.2.1 | CompTIA Network+ N10-009 Obj. 5.2 | N10-008 5.2

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

A split pair passes a basic continuity test because pin-to-pin connectivity is _____, but the two conductors are not properly _____ together, destroying interference cancellation. Only a _____ tester that measures crosstalk will catch it.

Question 2

Attenuation is measured in _____ and increases with cable _____. The maximum recommended run length for Cat6 copper cable is _____ meters total (including patch cables at each end).

Question 3

NEXT (Near-End Crosstalk) is caused by signal coupling between pairs at the _____ end of the cable. The most common installation cause is untwisting more than _____ of wire at the connector termination.

Question 4

Data cables running parallel to power cables should maintain at least _____ of separation. Where they must cross, they should cross at _____ to minimize electromagnetic coupling. If separation is impossible, use _____ cabling.

Question 5

A PoE port that repeatedly powers a device on and off is likely due to insufficient _____ on the switch. A device spec'd for 802.3at (PoE+, _____ watts) will fail on a switch that only supports 802.3af (_____ watts).

Question 6

Fiber optic bend radius violations occur when cable is bent tighter than the minimum specification, causing _____ to escape the fiber core (macrobending). The tool that makes these leaks visible is a _____.

Question 7

A switch port with a green link light but rising CRC error counts in `show interface` output indicates a _____ layer problem. Name three symptoms a user might report that actually trace back to CRC errors.

Question 8 — Real World Application

REAL WORLD: A newly installed patch panel in a wiring closet works for most ports, but four ports on one patch strip have poor throughput and high error rates at gigabit speeds, while basic link works fine at 100 Mbps. The cable runs are all under 15 meters. What is the most likely cause and how do you confirm it?

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/5-2-1-troubleshooting-cables-and-physical-layer/5-2-1-troubleshooting-cables-and-physical-layer-answer-key.pdf
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